

Department of Electrical & Electrical Engineering
BRAC University
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EEE283L

Digital Logic Design

Section: 03

Design Project : Smart voting System

Group -2

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Objective:

The goal of this project is to design and simulate a dual-purpose digital system for the BracU Optics and Photonics OPTICA student chapter elections. The system aims to provide a reliable and automated way to conduct a majority vote for 10 participants across three executive positions: President, Vice President, and Secretary. By utilizing combinational and arithmetic logic, the circuit identifies the winner based on the highest vote count and displays the result through visual indicators. Furthermore, the project includes the development of a digital stopwatch with a 0.1-second resolution to accurately monitor the election's duration. Both subsystems are designed with integrated control buttons Start, Stop, and Reset to ensure a smooth and professional administrative process.

Equipment:-

- Counter ICs (74161)
- Comparator ICs (7485)
- Logic Gate ICs (AND, OR, NOT)
- Logic toggle, probes
- 7-Segment Displays
- LED
- Push buttons and Toggles
- Jumper wires
- Digital Clock Source
- Proteus Professional 8 Software

Specification & Requirement:

Requirements:

- **Input Handling:** The system must feature 10 independent input switches to represent the 10 voters and 3 selection switches for the contestants.
- **Arithmetic Processing:** Magnitude comparators must be utilized to compare the bit-values of stored votes to determine the winner.
- **Timing Accuracy:** A stable clock pulse generator is required to ensure the stopwatch maintains an accurate 0.1-second resolution.
- **Synchronization:** The counter ICs must be synchronized to ensure that votes are registered and the timer increments without lag or error.
- **Status Indication:** The system requires a "Draw" detection logic to trigger an indicator if two or more contestants receive an equal number of votes.
- **User Control Interface:** Physical or logic-level toggles for Start , Stop and Reset
- **Power Stability:** A regulated +5V DC power supply requirement to maintain logic level integrity for the ICs.
- At least three primary LEDs for winners and two additional LEDs for Draw status .

Specification

The Smart Voting System is a dual-purpose digital device designed for conducting student organization elections while monitoring the administrative duration. The system is built using a combination of synchronous binary counters and magnitude comparators to handle a maximum of 10 voters and 3 distinct candidates. It operates on a stable +5V DC power supply to ensure the integrity of the logic levels across all integrated circuits (ICs). The architecture is divided into two primary subsystems: a voting logic unit and a digital stopwatch, both controlled by a unified interface consisting of Stop and Reset toggles.

Block diagram.

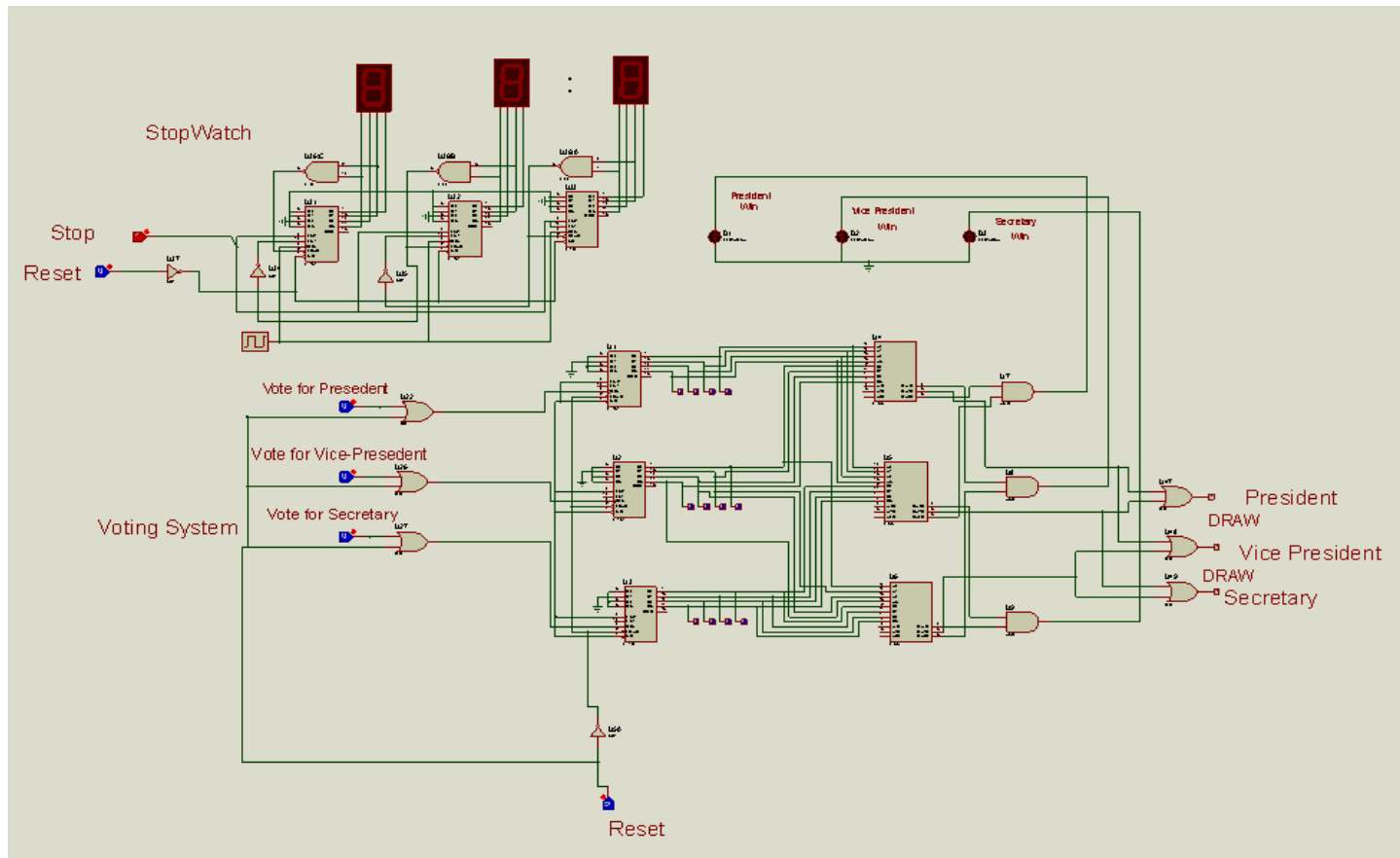
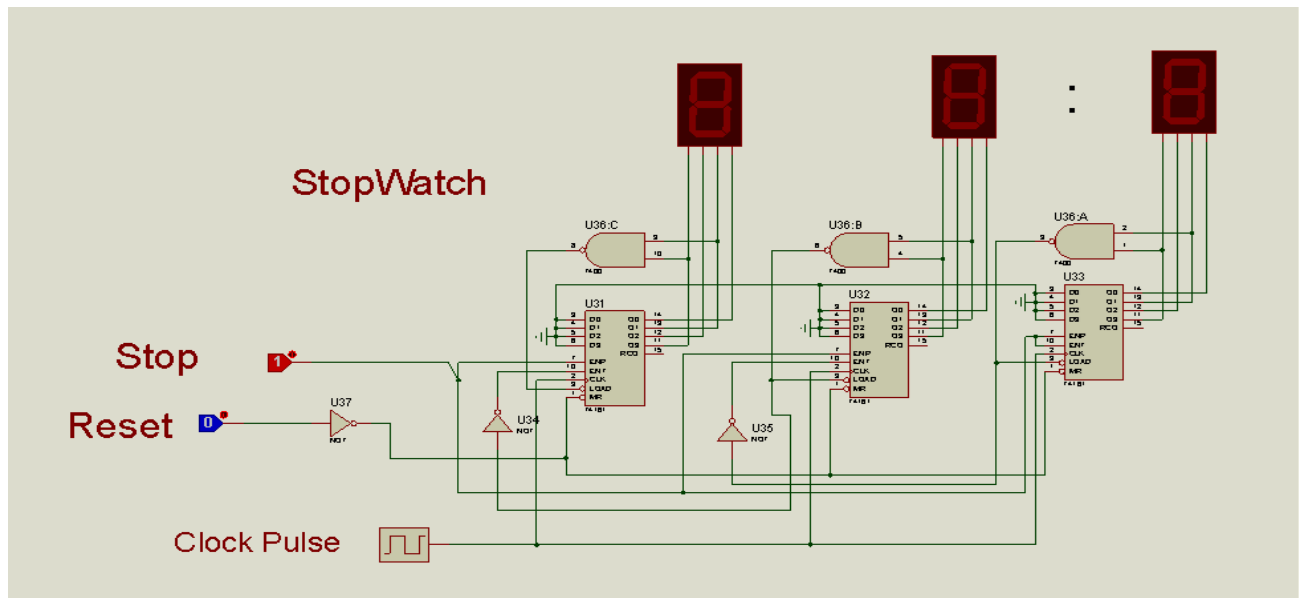
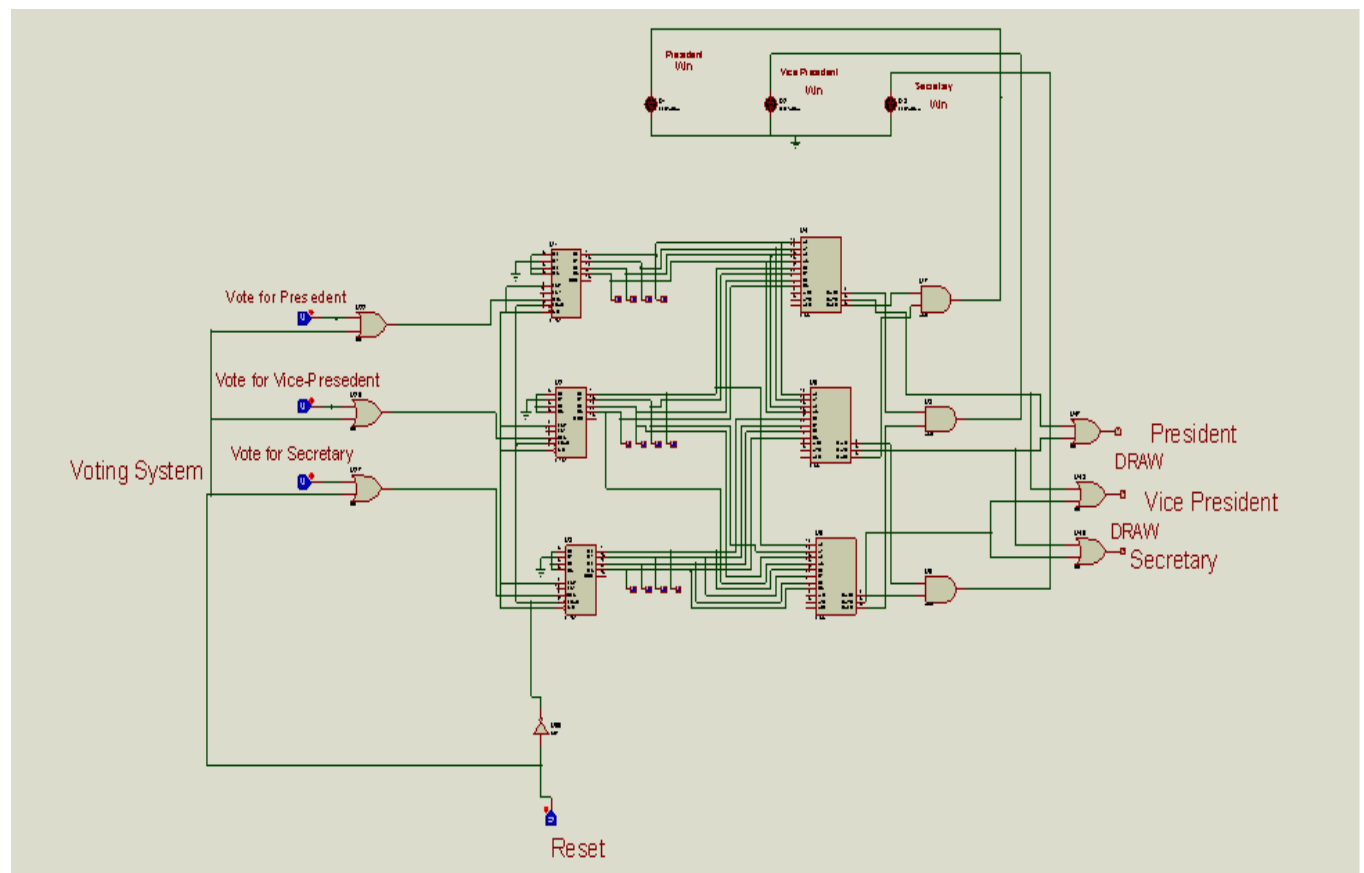


Figure: Full Circuit Diagram

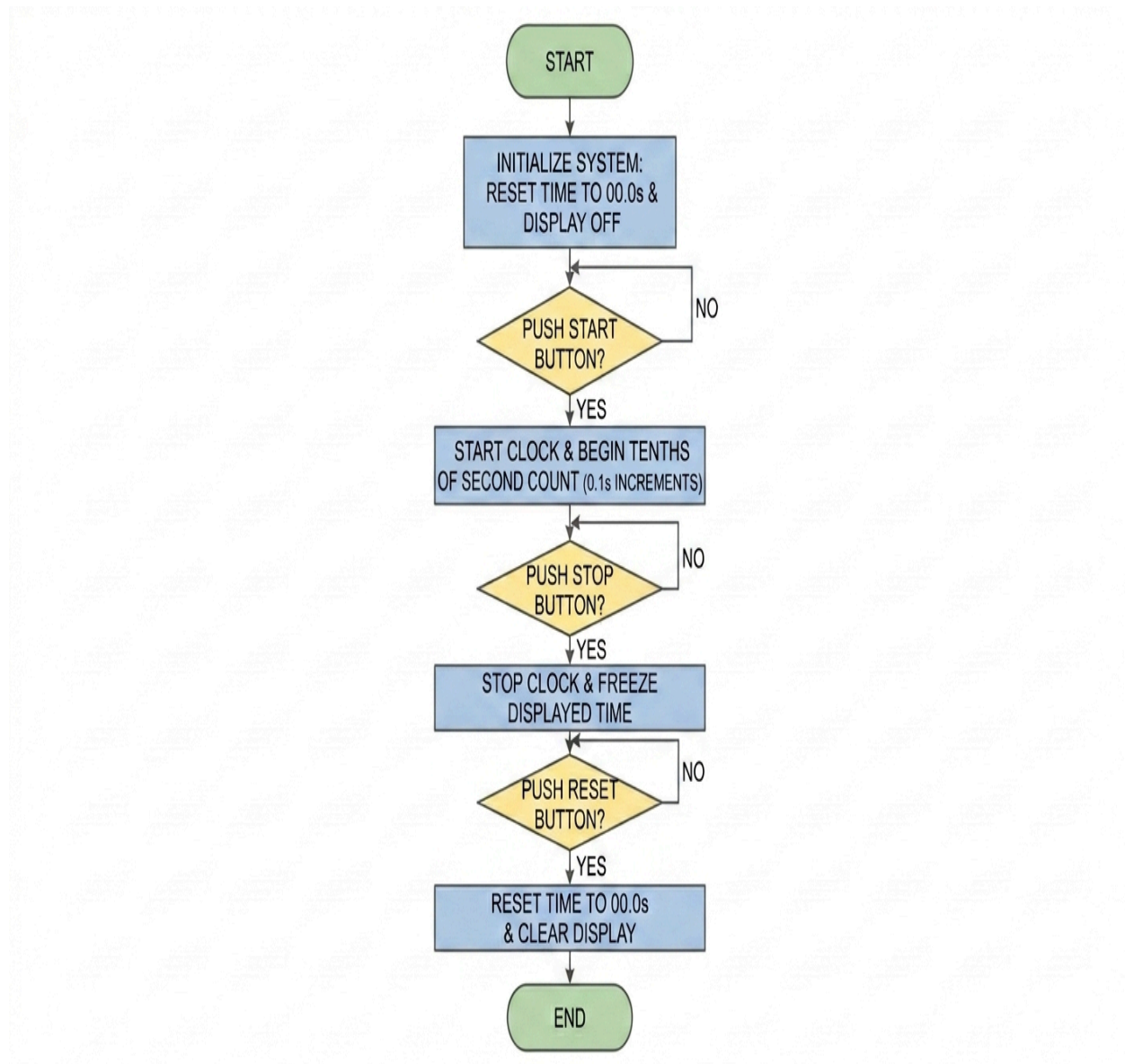
Stopwatch Subsystem:



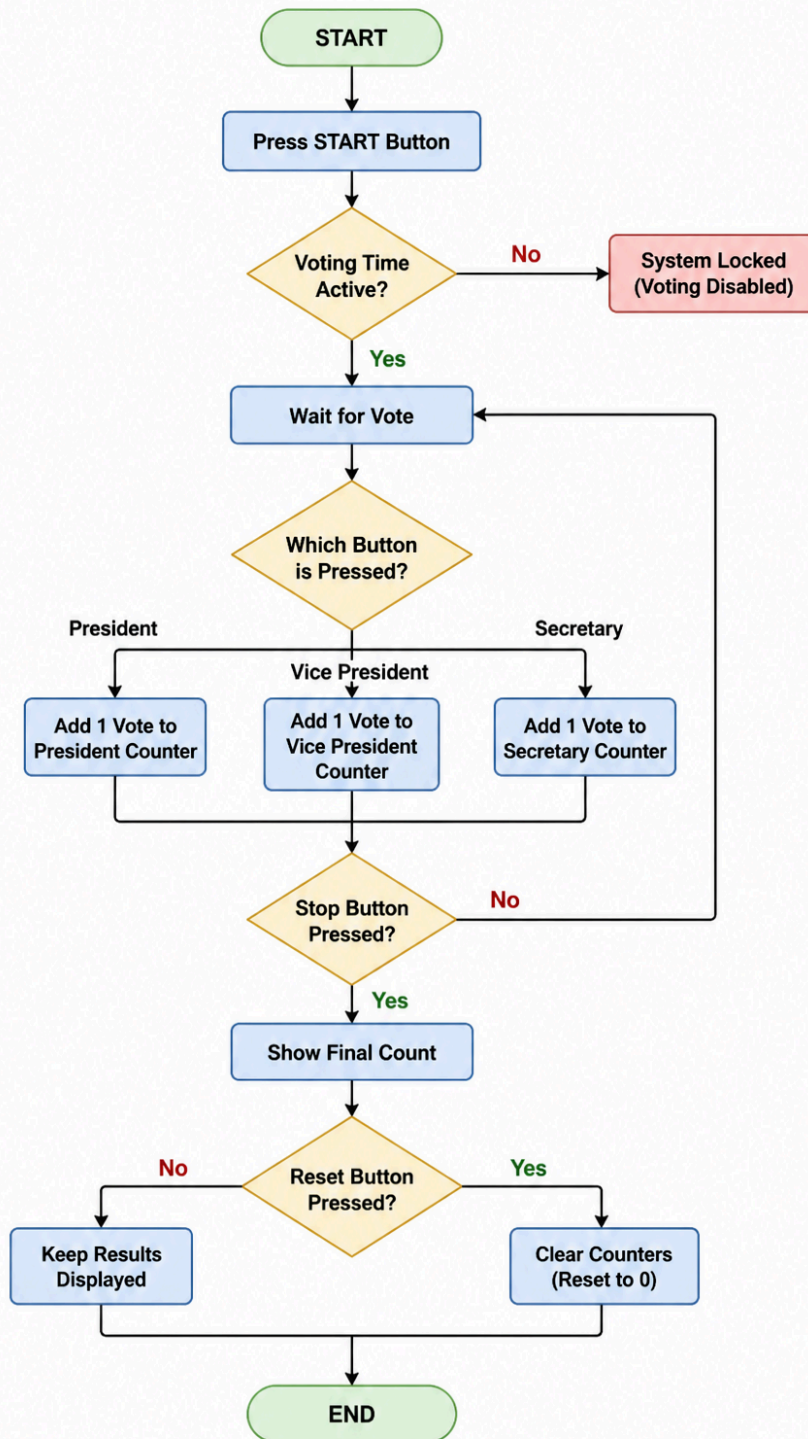
Voting Logic Subsystem



Flowchart:-
For Stopwatch



For Counter



Algorithm Development Procedure

Voting System Algorithm

1. Start the system.
2. Initialize all vote counters to 0.
3. Wait for voter input .
4. When a voter selects a candidate:
5. Add 1 vote to that candidate's counter.
6. Repeat until all votes are given.
7. Compare vote counts using comparator (7485):
 - If $A > B$ and $A > C \rightarrow A$ is winner
 - If $B > A$ and $B > C \rightarrow B$ is winner
 - If $C > A$ and $C > B \rightarrow C$ is winner
8. If two or more values are equal $\rightarrow$ Draw condition.
9. Display result using LED/LCD.
10. End.

Stopwatch Algorithm

1. Press Start to begin counting.
2. Clock pulse generates time (0.1 sec).
3. Counter (74161) increments:
 - Millisecond $\rightarrow$ Second
4. Press Stop for Pause counting.
5. Press Reset for All values become 0.
6. Display time on a 7-segment display.

Methodology

Overall System Design

The system is developed by dividing it into two main parts: the voting system and the stopwatch system. This approach helps to reduce complexity and allows each part to be designed and tested separately before combining them into a complete system. Both subsystems work together under a common control mechanism to ensure smooth operation.

Input and Control System

The input system is designed using switches and push buttons to make the system easy to operate. Each voter gives their vote using a push button and the system records it accordingly. To control the operation, We used:

Push buttons to take the input from participant

Reset → Clears all data

This simple control structure makes the system user-friendly and efficient.

Vote Counting Mechanism

Vote counting is implemented using synchronous binary counters (74161 IC), which operate on the rising edge of the clock signal. Each candidate is assigned a dedicated counter, ensuring parallel and independent vote storage.. This counter takes the input from the candidates and stores them and shows them in the display. Each counter has enough LED to show each contestant how many votes the contestant has.

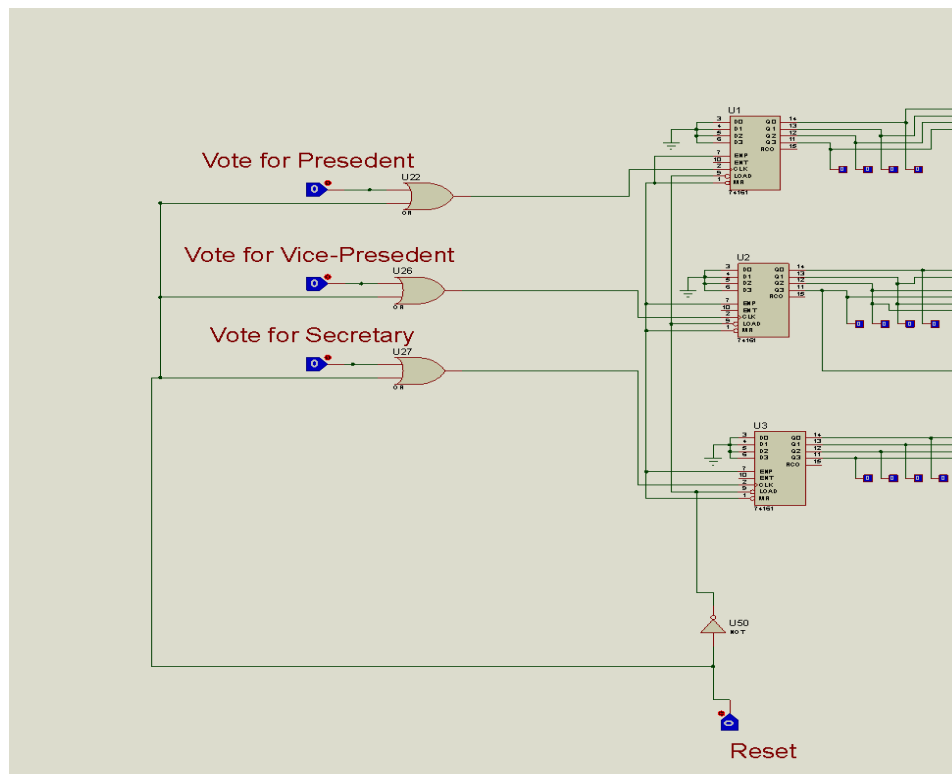


Figure: Counter part from voting system

Comparison and Decision Logic

The comparison process is performed using 4-bit magnitude comparators (7485 IC). These comparators generate three outputs:

- $A > B$
- $A = B$
- $A < B$

The outputs from multiple comparators are combined using logic gates to determine the final winner.

In here , we use 3 comparators ,

- The top one is mainly comparing the President and Vice-president.
- The middle one is comparing the President and Secretary.
- The bottom one is comparing the Vice-president and Secretary.

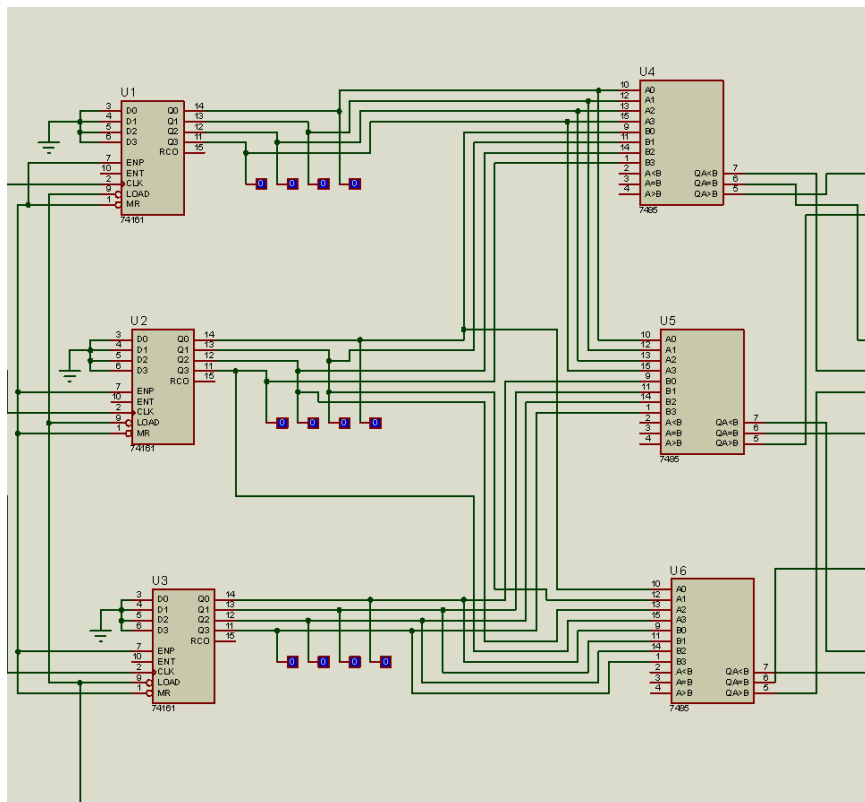


Figure: Comparison part from voting system

The logic for win:

- If $A > B$ and $A > C \rightarrow A$ is winner
- If $B > A$ and $B > C \rightarrow B$ is winner
- If $C > A$ and $C > B \rightarrow C$ is winner

Output Display System

The output system uses LEDs and display units to represent the final result. The logic outputs are directly connected to LED drivers, which convert digital signals into visual indications.

Stopwatch Design

The stopwatch subsystem is designed using a clock-driven sequential circuit. A clock generator produces a fixed frequency signal, which is divided to achieve a time resolution of 0.1 seconds.

The timing operation follows:

- Clock frequency division
- Cascaded counter design

Counters are connected in cascade form:

- First stage $\rightarrow$ counts milliseconds
- Second stage $\rightarrow$ counts seconds

All sequential components are synchronized using a common clock signal, ensuring deterministic behavior of the system. This eliminates race conditions and timing mismatches.

System Considerations

During design and implementation, several hardware-level factors are considered to improve system performance:

- Regulated +5V DC power supply ICs
- Proper grounding to avoid noise interference
- Debouncing of switches to prevent false triggering
- Clean clock signal generation

These considerations are important for both simulation accuracy and real hardware implementation.

Result:-

Truth Table

A>B	A=B	A<B	B>C	B=C	B<C	Winner
1	0	0	1	0	0	A
1	0	0	0	1	0	A
0	0	1	1	0	0	B
0	0	1	0	0	1	C
0	1	0	0	1	0	Draw (A=B=C)
1	0	0	0	0	1	A
0	0	1	0	1	0	Draw (B = C)
0	1	0	1	0	0	Draw (A = B)

Test Case:

For President Win & (Vice president = Secretary)

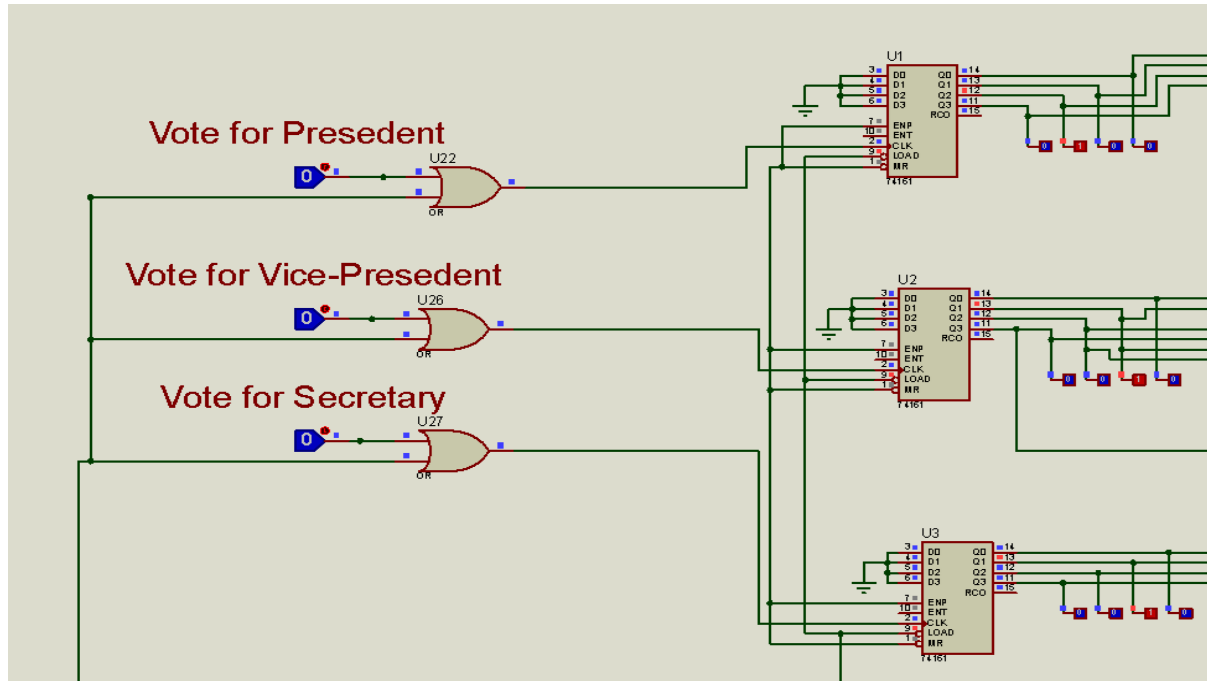


Figure: Input votes[P=4,VP=2,S=2]

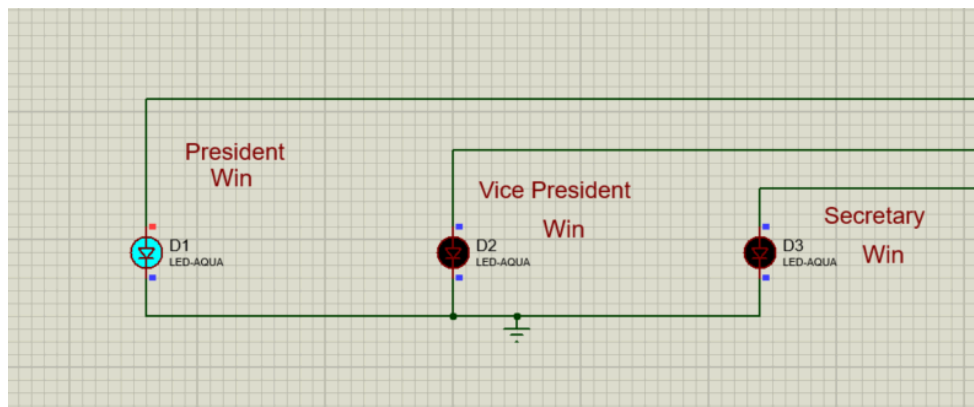


Figure: President Win

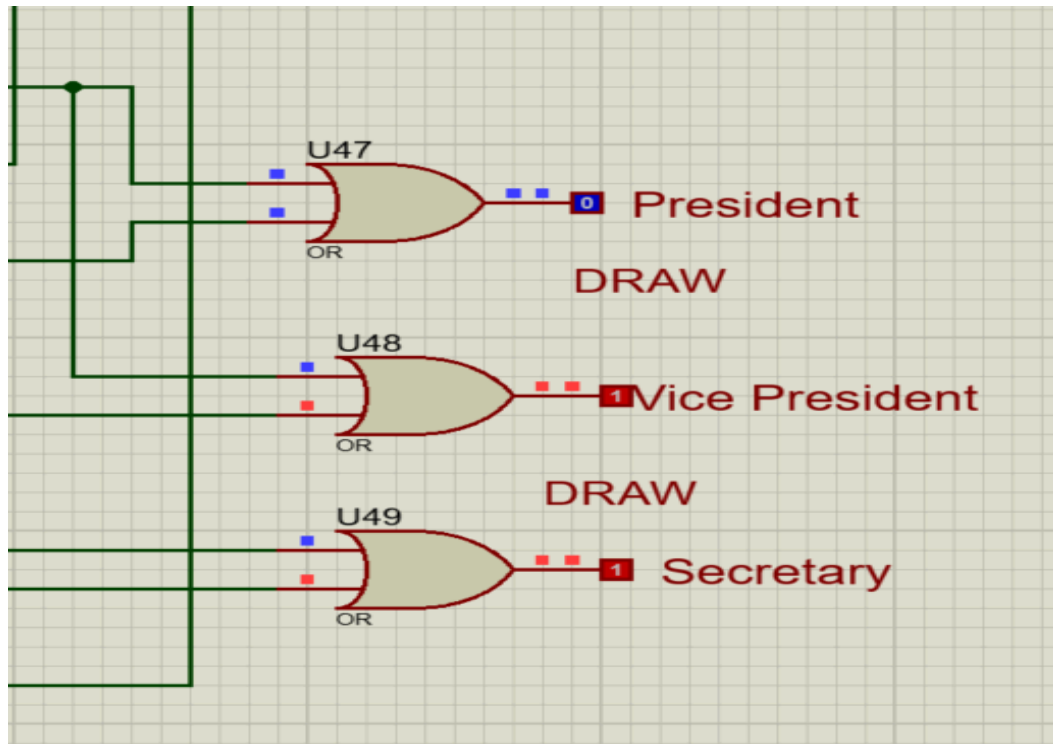


Figure: Vice president & Secretary

For contestants are draw,

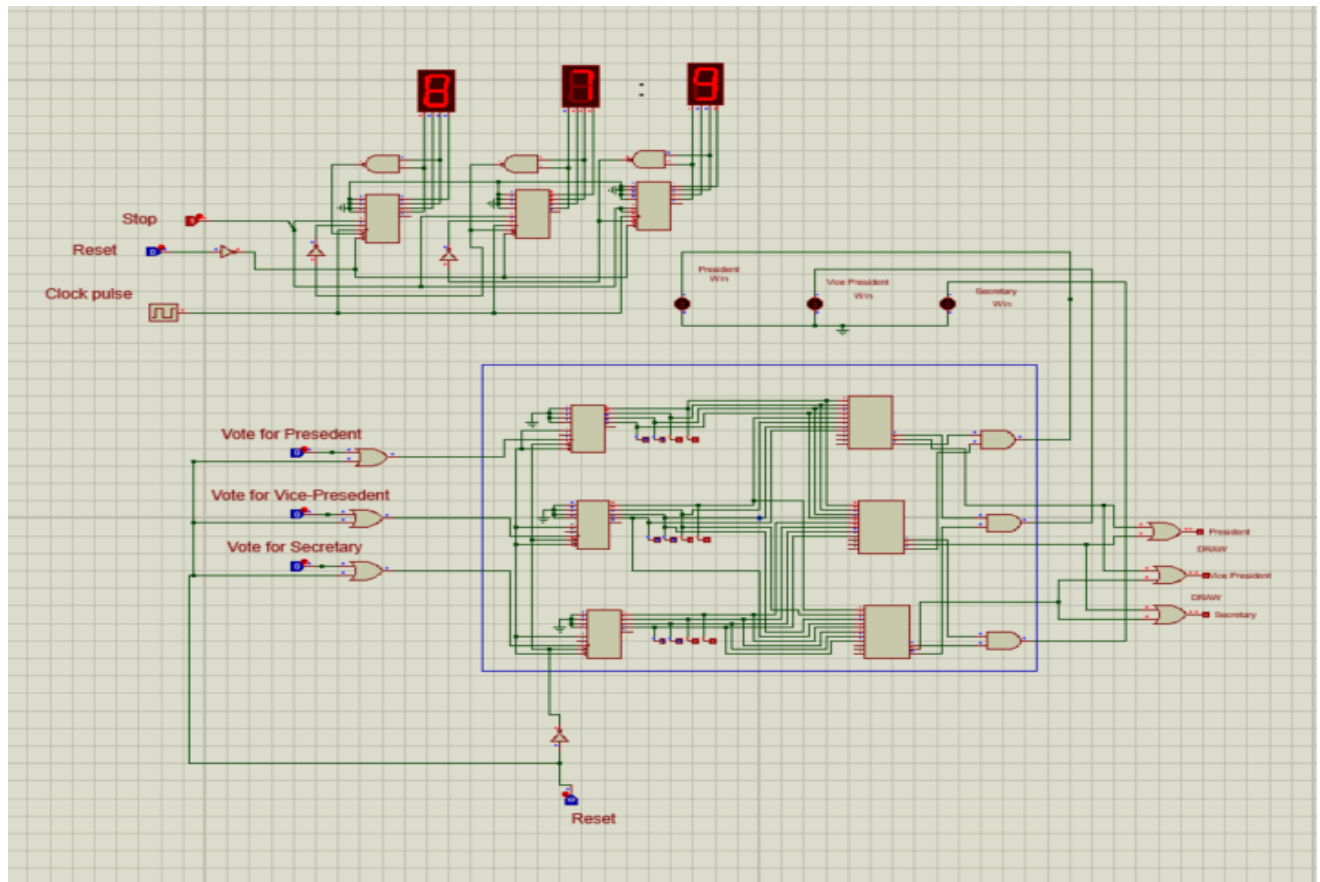


Figure: stopwatch is on and every contestant is draw

Hardware Part

Voting System Part

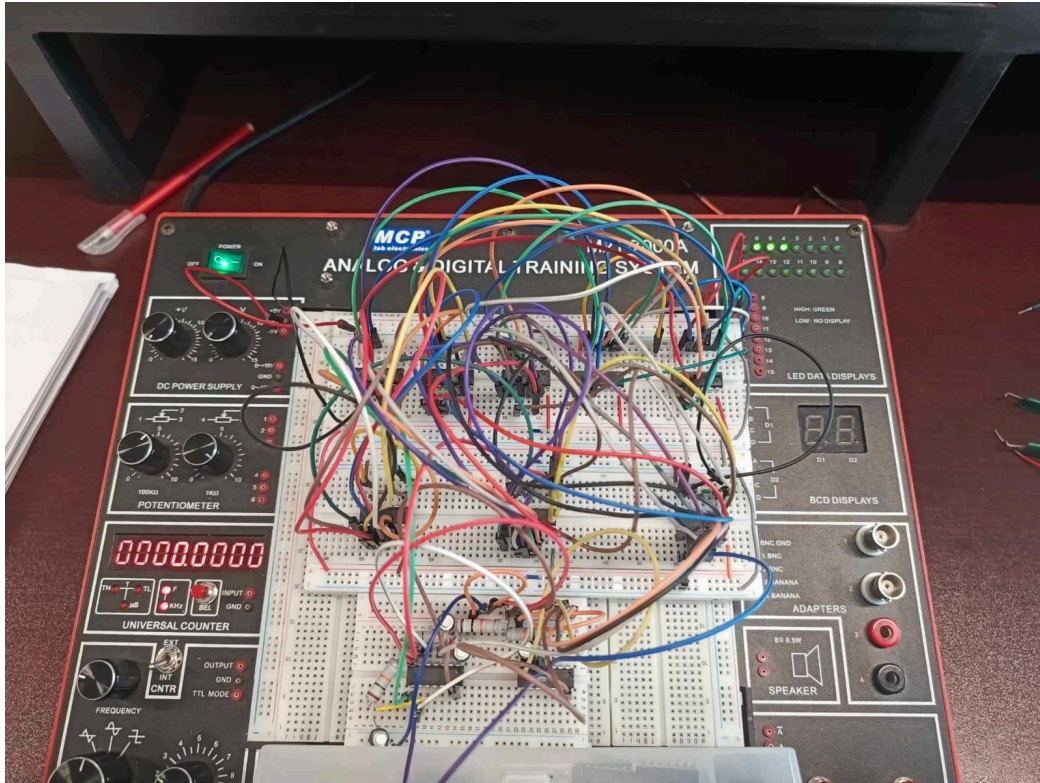


Figure: Initial situation all vote is zero ,so all draw.

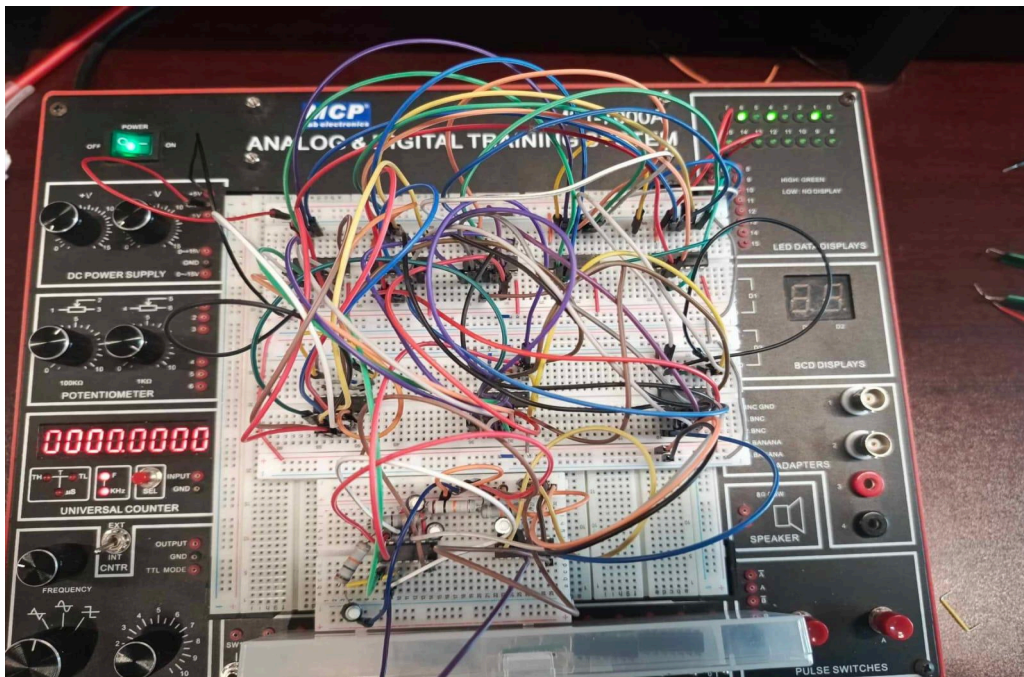


Figure: B Win and A,C is draw

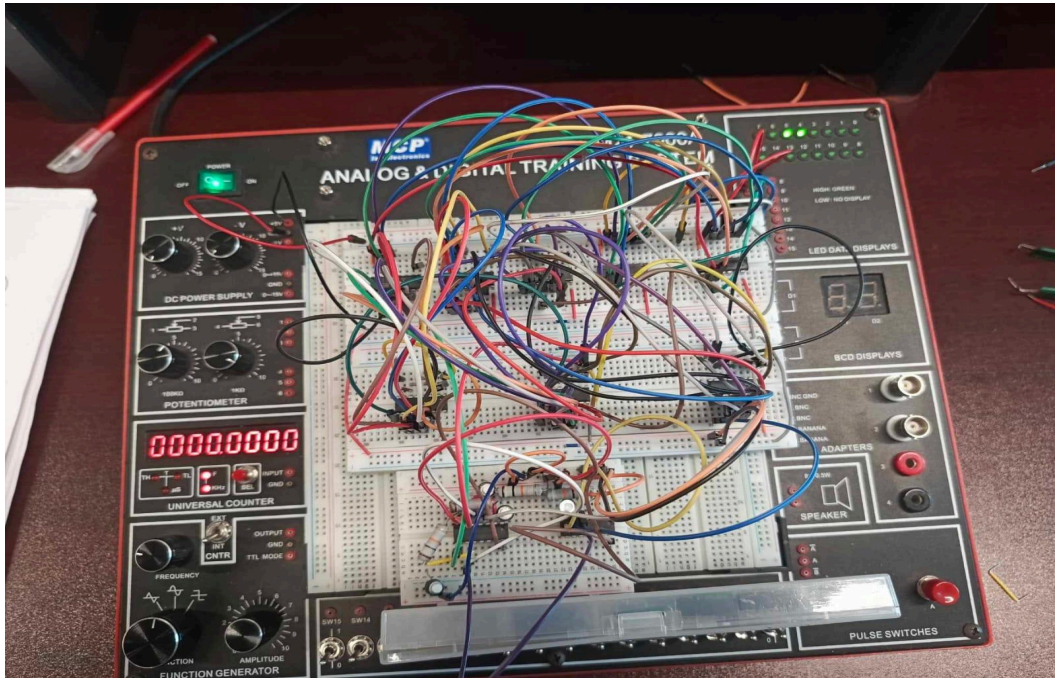


Figure: B and C is draw , No winner

Stopwatch System part

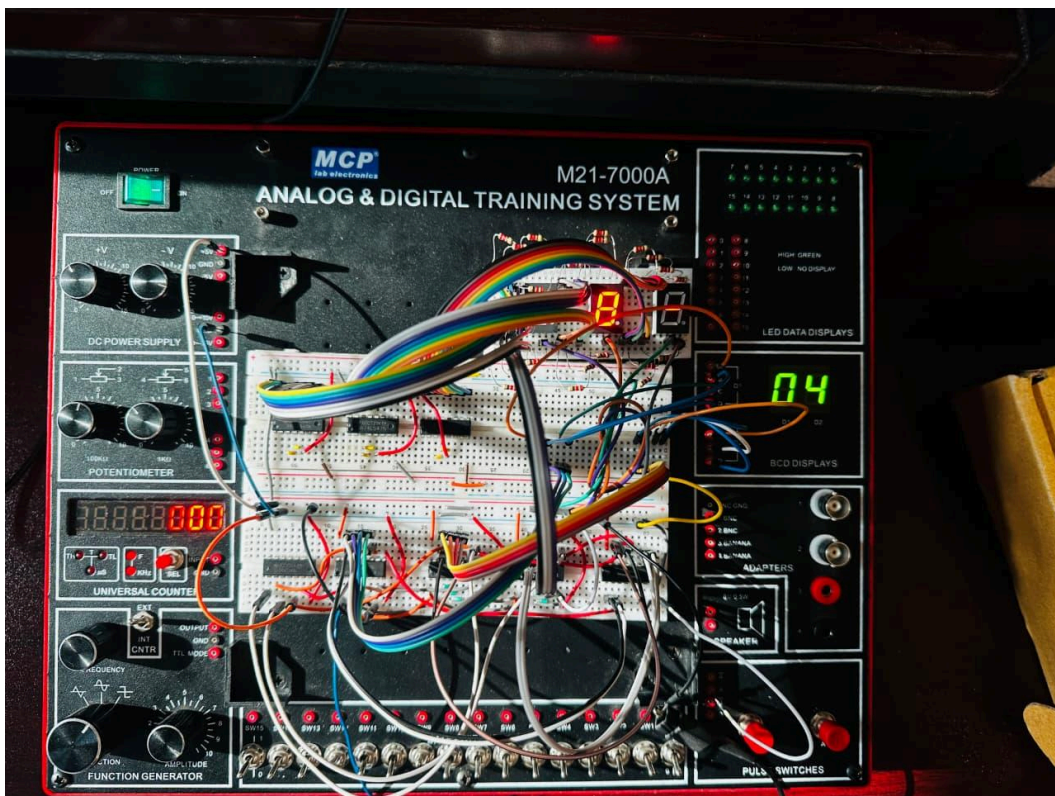


Figure: Stopwatch running for 10 Hz

Discussion:

The Smart Voting System successfully integrates combinational and sequential logic to automate election management for the BracU Optics and Photonics student chapter. By dividing the project into a voting unit and a stopwatch subsystem, the design remains organized and allows each part to be tested independently. The voting logic uses 74161 counters to record votes in parallel, while 7485 magnitude comparators compare these totals to determine the winner based on the highest count. To ensure the process is professional and accurate, the system includes a "Draw" detection logic and a stopwatch with 0.1-second resolution to track the election duration. Technical factors like switch debouncing, proper grounding, and a regulated +5V power supply were used to prevent errors and noise interference. Overall, the project provides a reliable, user-friendly digital solution that ensures a smooth administrative process.